

RAMCO INSTITUTE OF TECHNOLOGY
Department of Electronics and Communication Engineering
Academic Year: 2019- 2020 (Odd Semester)

Innovative Practices Description

Degree, Semester & Branch: V Semester B.E. ECE A

Course Code & Title: EC8551 Communication Networks

Name of the Faculty member: Mr.K.Ragavan

Name of the Topic: OSI layer

Name of the Innovative Practice: One Minute Paper

Date & Time: 10.07.2019 & 1 Minute

Description:

A “one-minute paper” may be defined as a very short, in-class writing activity (taking one-minute or less to complete) in response to an instructor-posed question, which prompts students to reflect on the day’s lesson and provides the instructor with useful feedback.

Minute papers can provide a “conceptual bridge” between successive class periods. For instance, at the beginning of class, a quick review of student responses to a minute paper answered at the end of a previous class can provide an effective segue between successive class sessions.

Learning Outcomes:

1. The students will be able to enumerate the layers of the OSI model.
2. The students will be able to understand and explain the functions of layers in OSI Model.
3. The students can be easily identify the addresses used in seven layers in the OSI model.

Justification for choosing the topic using One Minute Paper activity:

The present study attempts to introduce one minute paper as an innovative learning method. A one minute paper is a quick way of gathering information on how well our students are meeting their learning objectives. One minute paper is like simple intervention and it can make a significant improvement in the knowledge of engineering students and help them to develop interest in OSI Model.

Implementation (Plan & Execution) with Proof:

One minute paper is one of the assessment tool with the following characteristics:

1. Specific question was asked, specific answer are required; either in a word or in a couple of sentences.
2. The correct answers are pre-determined, not affected by students’ opinionating or reasoning.
3. The exercise is conducted in a controlled, monitored framework (a classroom), usually within a short time limit.

In my subject one minute paper was conducted for Unit-I OSI Model at the completion of the topic.

Innovative Teaching Method Execution

OSI layer : One-Minute Paper



Reflective Critique:

- **Benefits:**

The one minute paper was conducted from OSI Model. The answers reflect the students' understanding, ability to recall. Students do not feel that they are being judged by a different measure than that of any other student. After the quiz, students gain knowledge by learning the correct answers to the question.

- **Challenges:**

Few students take more than the time given for this activity

Course Outcome	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	2	2	1	1					2	2		

References:

- Behrouz A. Forouzan, “Data communication and Networking”, Fifth Edition, Tata McGraw – Hill, 2013

CO1: The students will be able to describe the concepts of the network fundamentals and different layers.

RAMCO INSTITUTE OF TECHNOLOGY
Department of Electronics and Communication Engineering
Academic Year: 2019- 2020 (Odd Semester)

Innovative Practices Description

Unit-II Media Access & Internetworking

Degree, Semester & Branch: V Semester B.E. ECE A

Course Code & Title: EC8551 Communication Networks

Name of the Faculty member: Mr.K.Ragavan

Name of the Topic: Ethernet (802.3) - RJ45

Name of the Innovative Practice: Hardware Demonstration

Date & Duration: 19.07. 2019 & 30 Minutes

ICT Tool Used: RJ45 Cable

Description:

Hardware demonstrations can be used in lectures to demonstrate the application of a concept, a skill, or to act out a process. It creates opportunities for students to reflect and analyze the process. demonstration has the potential advantage of being ‘better, more visible, clearer and with more impact than a class experiment’ .This gives them the opportunity to develop skills in handling and using equipment, making decisions, collecting data and actively thinking about what they are doing and learning.

Learning Outcome:

- The students can acquire better knowledge about RJ 45 Cable used in Ethernet.
- The students understood the difference between various Ethernet cables used in Real Time applications.

Justification for choosing the topic using Hardware Demonstration activity:

In teaching through demonstration, students are set up to potentially conceptualize class material more effectively. Relating and demonstrating through real-life situations will make the material easy to understand and easy to learn. It will spark their interest and get the students excited and involved.

Implementation (Plan & Execution) with Proof:

Innovative Teaching Method Execution

Ethernet :- RJ45 Hardware Demonstration



Reflective Critique:

- **Benefits:**

The benefits of demonstration as a teaching method are a better learning experience in the classroom for students; the generation interest in the subject; help in developing the spirit of inquiry; students cooperating in the teaching-learning process; and the process of learning becoming permanent in the student's life.

- **Challenges:** Time management is complicated one to shown the cable to the students individually

Course Outcome	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO2	2	2	1	1					2	1		

Reference:

- Behrouz A. Forouzan, “Data communication and Networking”, Fifth Edition, Tata McGraw – Hill, 2013

CO2: The students will be able to identify the components required to build different types of networks and internetworking protocols.

RAMCO INSTITUTE OF TECHNOLOGY
Department of Electronics and Communication Engineering
Academic Year: 2019- 2020 (Odd Semester)

Innovative Practices Description

Unit-III Routing

Degree, Semester & Branch: V Semester B.E. ECE A

Course Code & Title: EC8551 Communication Networks

Name of the Faculty member: Mr.K.Ragavan

Name of the Topic: Transaction from IPv4 to IPv6

Name of the Innovative Practice: Mind Map

Date & Duration: 30.08.2019 & 10 Minutes

ICT Tool Used: Black Board

Description:

A mind map is a diagram used to visually organize information. A mind map is hierarchical and shows relationships among pieces of the whole. Mind mapping in a learning environment which requires an enhanced way of thinking, learning and presenting. Mind Mapping enhances the learning capacity in terms of number of ideas generated as well as improves presentation focus. Mind maps are used to generate, visualize, structure and classify ideas. Mind maps go under variety of names. They are also known as concept maps, semantic mapping, knowledge mapping, think links, graphic organizers or cognitive maps.

Learning Outcome:

- The students can acquire better knowledge about different types of intra and interdomain routing .
- The students will be able to visualize the concepts and organize information.

Justification for choosing the topic using Mind Map activity:

Mind mapping is a creative way to set goals, solve problems and design action plans. It quickly records ideas in a free-form way. When groups use mind mapping, the thoughts of each participant easily trigger ideas in others. This dynamic group interaction encourages breaking free of old patterns to uncover new and innovative approaches.

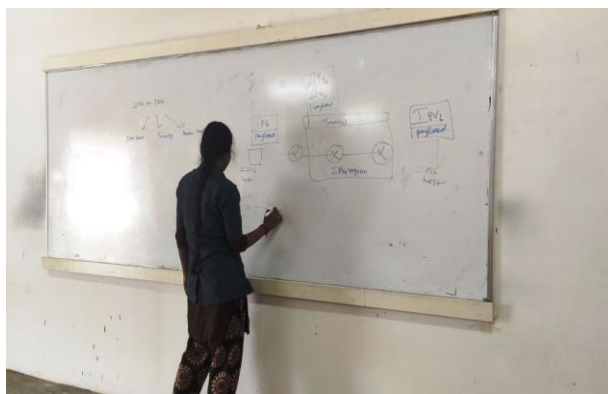
Implementation (Plan & Execution) with Proof:

Mind mapping activities require students to actively engage in their learning, often by connecting their prior knowledge to new information. When creating a mind map, a student frequently interacts with a textbook, notes from class, an instructor, classmate, or study group.

In my subject Mind map was conducted for Unit-III Routing at the completion of unit. The students are voluntarily drawn the map. The sample copy of the mind map was done by the students using white board was shown below.

Innovative Teaching Method Execution

Transaction from IPv4 to IPv6 – Mind Map



Reflective Critique:

- Benefits:**

The benefits of Mind Maps are a great way for students to make notes on all of the information they receive. It helps the students to note down only the most important information using key words, and then make connections between facts and ideas visually – keeping all of your topic thoughts together on one sheet. It made key note making easier to students, as it reduces pages of notes into one single side of paper. Also mind map made slow learners to remember the information more quickly.

- Challenges:**

Normally teachers will give longer explanations in the notes section of the topic. At the completion of the unit, teacher will revise the unit. The students are called randomly to draw the map to indicate relationships between the topics in Routing. I planned the activity for 5 minutes only. But in real scenario it takes 10 minutes to complete the whole topics in Unit-III.

Course Outcome	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO3	2	1		1				1	2	1		

Reference:

- Behrouz A. Forouzan, “Data communication and Networking”, Fifth Edition, Tata McGraw – Hill, 2013

CO3: The students will be able to The students will be able to apply the concept of various protocols in routing and multicasting.

RAMCO INSTITUTE OF TECHNOLOGY
Department of Electronics and Communication Engineering
Academic Year: 2019- 2020 (Odd Semester)

Innovative Practices Description

Unit-IV Transport Layer

Degree, Semester & Branch: V Semester B.E. ECE A

Course Code & Title: EC8551 Communication Networks

Name of the Faculty member: Mr.K.Ragavan

Name of the Topic: Transport layer

Name of the Innovative Practice: Quiz

Date & Duration: 19.09.2019

ICT Tool Used: Online Google Form

Description:

A quiz can be defined as a game or brain teaser to test knowledge. It can contain an element of competition when participants play against each other to get the highest score, which makes helps participants become more engaged. The quiz consists of a large variety of Question types, including multiple choice, true-false, short answer and drag and drop images and text. These questions are kept in the Question bank and can be re-used in different quizzes.

Learning Outcome:

- The students will be able to understand the different functionalities of Transport Layer.

Justification for choosing the topic using Quiz activity:

The present study attempts to introduce quiz as an innovative learning method. A quiz is a quick way of gathering information on how well our students are meeting their learning objectives. Quiz is like simple intervention and it can make a significant improvement in the knowledge of engineering students and help them to develop interest in Combinational Logic Circuits.

Implementation (Plan & Execution) with Proof:

Quiz is one of the assessment tool with the following characteristics:

1. Specific questions are asked, specific answers are required; either in a word or in a couple of sentences.
2. The correct answers are pre-determined, not affected by students opinionating or reasoning.
3. The exercise is conducted in a controlled, monitored framework (a classroom or an online quiz area), usually within a short time limit.

In my subject quiz was conducted for Unit-IV Transport Layer at the completion of unit through Google forms. 20 questions are posted to the students College email id using Google forms. The students are answered the Questions within a week. The screen shot of the students marks, questions and answers are shown below.

Innovative Teaching Method Execution												
Transport layer - Quiz												
Blank Quiz (Responses) ☆ File Edit View Insert Format Data Tools Form Add-ons Help 100% \$ % .00 .00 123 Default (Ari... 10 B I A												
fx	Timestamp											
	B	C	D	E	F	G	H					
1	Score	Register Number	Name of the student	Transport layer protocols	2. Which one of the follow	3. User datagram protoco	4. Transmission control pi	5. TCP				
2	9 / 20	953617106005	Aishwarya.B	b) process to process con	c) Both TCP and UDP	b) it sends data as a stre	a) connection oriented pr	a) Pacl				
3	18 / 20	953617106052	S.MAHALAKSHMI	b) process to process con	a) TCP	a) all UDP packets are tre	d) all of the mentioned	c) Segr				
4	16 / 20	953617106050	R.LEON MOBIL BRINGA	b) process to process con	c) Both TCP and UDP	c) it is received in the sar	d) all of the mentioned	c) Segr				
5	16 / 20	953617106047	lakshmikantham	b) process to process con	c) Both TCP and UDP	b) it sends data as a stre	a) connection oriented pr	c) Segr				
6	3 / 20	953617106021	dhanushya.r	b) process to process con	b) UDP	b) it sends data as a stre	b) uses a three way hand	b) Buffi				
7	20 / 20	953617106024	T.DhivyaMuthachi	b) process to process con	c) Both TCP and UDP	a) all UDP packets are tre	d) all of the mentioned	c) Segr				
8	5 / 20	953617106007	R.Amaravathi	c) node to node communi	b) UDP	b) it sends data as a stre	a) connection oriented pr	b) Buffi				
9	15 / 20	953617106016	M.Aysha Shahana	b) process to process con	c) Both TCP and UDP	d) none of the mentioned	d) all of the mentioned	c) Segr				
10	14 / 20	953617106011	R.Anushya	c) node to node communi	c) Both TCP and UDP	b) it sends data as a stre	d) all of the mentioned	c) Segr				
11	12 / 20	953617106031	JaiHarish	b) process to process con	c) Both TCP and UDP	a) all UDP packets are tre	d) all of the mentioned	c) Segr				
12	10 / 20	953617106029	A.Harnee	b) process to process con	c) Both TCP and UDP	d) none of the mentioned	a) connection oriented pr	c) Segr				

Reflective Critique:

• Benefits:

The quiz was conducted with multiple choices. The answers reflect the students' understanding, ability to recall, as well as the ability to apply a concept learnt in the class. Fairness is implemented in the objective nature of the quiz, i.e students do not feel that they are being judged by a different measure than that of any other student. Students can display their knowledge by overcoming certain difficulties faced by them by a paper assignment for eg. Language skills, paper anxiety, viva anxiety, word limitations etc. After the quiz, students gain knowledge by learning the correct answers to any questions they missed, can highlight the areas requiring improvement or reconsider their study habits.

• Challenges:

Few students take more than the time given for this activity

Course Outcome	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO4	2	2	1	1					2	1		

Reference:

- Behrouz A. Forouzan, "Data communication and Networking", Fifth Edition, Tata McGraw – Hill, 2013

CO4: The students will be able to explain the flow of information from one node to another in the networks.

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Innovative Practices Description

Unit-V Application Layer

Degree, Semester & Branch: V Semester B.E. ECE A

Course Code & Title: EC8551 Communication Networks

Name of the Faculty member: Mr.K.Ragavan

Name of the Topic: Need for Cryptography & Network security

Name of the Innovative Practice: Video Clip

Date & Duration: 03.10.2019

ICT Tool Used: LCD Projector

Description:

Use of animation in special education has gained great attention recently. Student engagement and understanding of materials is given more emphasis in today's education over spoon feeding the facts. Therefore, using black-boards or the typical lecture methods are not adequate to teach science and other related subjects. The evolution of a process can be conveyed better with animation videos. It can also be videos of demonstration of an idea or an application side of a theory or an interview with a scientist, tutorial by a subject expert and more.

Learning Outcome:

- The students will be able to easily identify the different types of keys used in Cryptography and network security
- The students will be able to explain the application of Cryptography and network security.

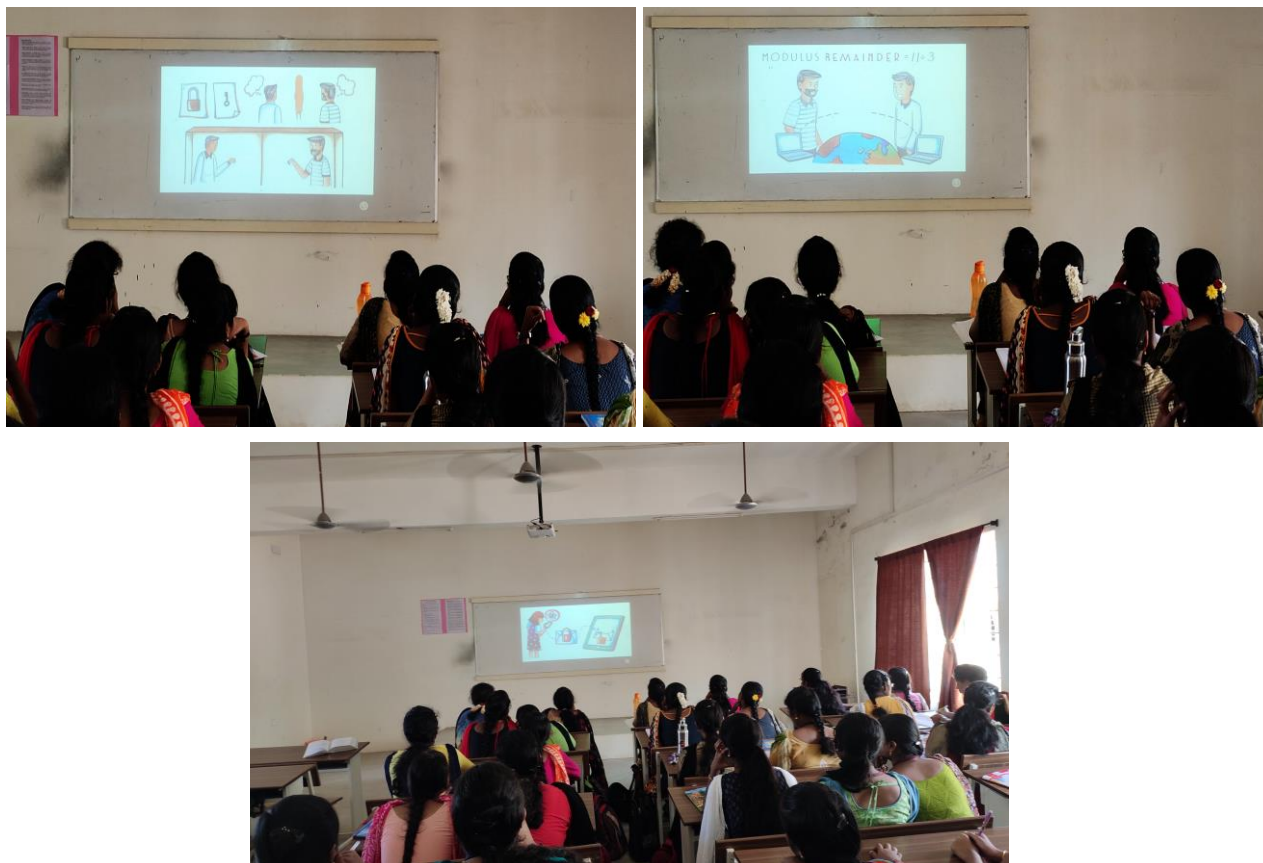
Justification for choosing the topic using Video Clip activity:

Video is one of the most engaging mediums of the modern day. Animation videos are one of the digital tools for classroom. It enables students to apply Imagination & think individually about a topic or answer to a question. It helps to promote creativity and motivate the students to do mini projects using memory devices

Implementation (Plan & Execution) with Proof:

Innovative Teaching Method Execution

Need for Cryptography & Network security – Video Clip



Reflective Critique:

• Benefits:

- The students are more interested to watch animated video, apart from theory concept.
- Using this activity strengthen the students talent and skill set.
- It helps the students to communicate stories, ideas and concepts in a creative and original way.

• Challenges:

It takes more time to choose relevant animated videos for easily understanding the concepts of different types of keys used in cryptography and various network security techniques.

Course Outcome	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO5	2	2	1	1					2	1		

Reference:

- Behrouz A. Forouzan, “Data communication and Networking”, Fifth Edition, Tata McGraw – Hill, 2013
- www.youtube.com

CO5: The students will be able to analyze the operations of various application layer protocols such as WWW, HTTP, and DNS.

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Innovative Practices Description

Unit-V Application Layer

Degree, Semester & Branch: V Semester B.E. ECE A

Course Code & Title: EC8551 Communication Networks

Name of the Faculty member: Mr.K.Ragavan

Name of the Topic: DNS, Firewall

Name of the Innovative Practice: Hardware Demonstration

Date & Duration: 26.09.2019 & 40 Minutes

ICT Tool Used: RIT-Data Centre

Description:

Hardware demonstrations can be used in lectures to demonstrate the application of a concept, a skill, or to act out a process. It creates opportunities for students to reflect and analyze the process. Demonstration has the potential advantage of being ‘better, more visible, clearer and with more impact than a class experiment’. This gives them the opportunity to develop skills in handling and using equipment, making decisions, collecting data and actively thinking about what they are doing and learning.

Learning Outcome:

- The students can acquire better knowledge about Firewalls and Domain Name Server.
- The students understood the difference between DNS used in Real Time applications.

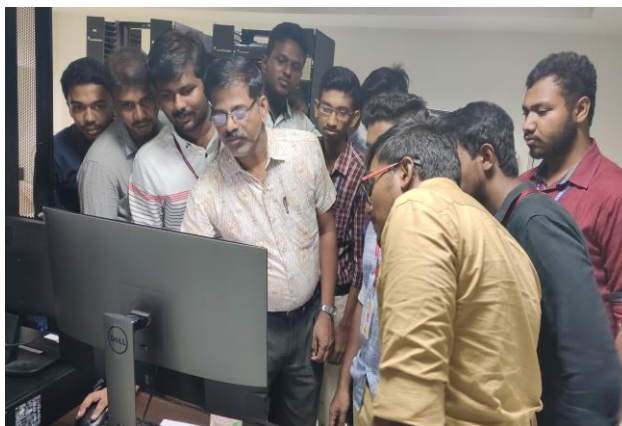
Justification for choosing the topic using Hardware Demonstration activity:

In teaching through demonstration, students are set up to potentially conceptualize class material more effectively. Relating and demonstrating through real-life situations will make the material easy to understand and easy to learn. It will spark their interest and get the students excited and involved.

Implementation (Plan & Execution) with Proof:

Innovative Teaching Method Execution

DNS, Firewall - Hardware Demonstration



Reflective Critique:

• Benefits:

The benefits of demonstration as a teaching method are a better learning experience in the classroom for students; the generation interest in the subject; help in developing the spirit of inquiry; students cooperating in the teaching-learning process; and the process of learning becoming permanent in the student's life.

• Challenges:

Time management is complicated one to visit and explain about all the equipments available in the data centre to the students.

Course Outcome	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO5	2	2	1	1					2	1		

CO5: The students will be able to analyze the operations of various application layer protocols such as WWW, HTTP, and DNS.